

TASK 2

Implement symmetric block cipher encryption and decryption using DES algorithm in C/JAVA

Aim: To write a Java Program on DES algorithm

Program:

```
import java.security.*;
import java.io.*;
import javax.crypto.*;
import java.util.*;

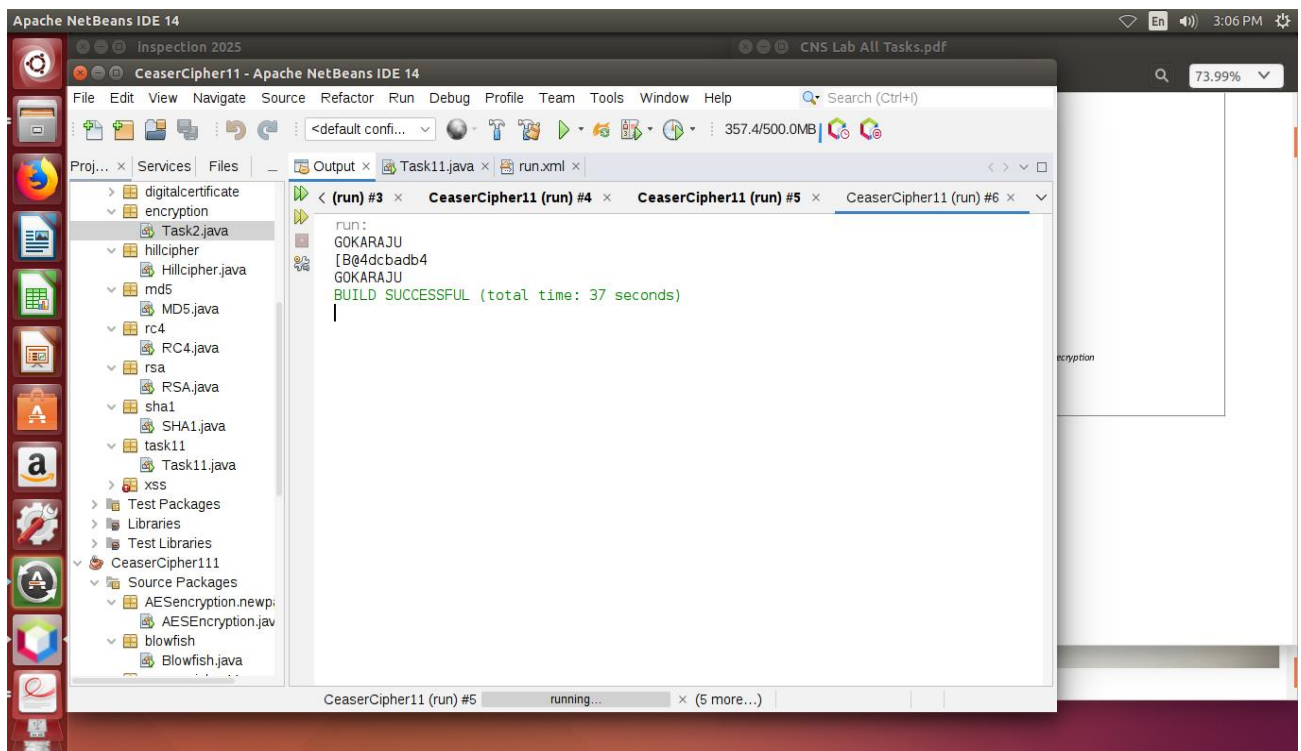
//DES
public class Task2 {
    static Scanner s = new
    Scanner(System.in); private static Cipher
    cipher;
    private static byte[] encrypt(String data) throws Exception{
        byte[] encrypted= cipher.doFinal(data.getBytes());
        System.out.println(encrypted);
        return encrypted;
    }
    private static byte[] decrypt(byte[] data) throws Exception{
        byte[] decrypted= cipher.doFinal(data);
        System.out.println(new
        String(decrypted)); return decrypted;
    }
    public static void main(String[] args)
    { try {
        String data = s.nextLine();
        KeyGenerator keygen =
        KeyGenerator.getInstance("DES");
        SecretKey key = keygen.generateKey();
        cipher=Cipher.getInstance("DES/ECB/PKCS5Paddin";
        cipher.init(Cipher.ENCRYPT_MODE,key);
        byte[] encrypted = encrypt(data);
```

```

        cipher.init(Cipher.DECRYPT_MODE, key);
        decrypt(encrypted);
    } catch (Exception e)
    {
        e.printStackTrace();
    }
}
}
}

```

OUTPUT:



VIVA QUESTIONS

1. What is a block cipher, and how does it differ from a stream cipher?
2. Can you explain the working of a common block cipher, such as AES or DES?
3. What are the different modes of operation used in block ciphers, and why are they important?
4. How does key length impact the security of a block cipher?